plotting the estimated index value against 100 per cent

---CHUNK---

SOURCE: MEPC.212(63).pdf
i
AEi ME
V
PP
Value IndexEstimated

DWT % 70
215190
1144 . 3 1

.3 parameters a and c for containerships in Table 2 of regulation 21 of MARPOL Annex VI are determined by plotting the estimated index value against 100 per cent deadweight i.e. a=174.22 and c=0.201 were determined.

.4 required EEDI for a new containership is calculated using 100 per cent deadweight as:

Required EEDI = (1-X/100) · a · 100% deadweight c

Where X is the reduction factor (in percentage) in accordance with Table 1 in regulation 21 of MARPOL Annex VI relating to the applicable phase and size of new containership.

.4 Deadweight means the difference in tonnes between the displacement of a ship in water of relative density of 1,025 kg/m 3 at the summer load draught and the lightweight of the ship. The summer load draught should be taken as the maximum summer draught as certified in the stability booklet approved by the Administration or an organization recognized by it.
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 5

---CHUNK---

SOURCE: MEPC.212(63).pdf
RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 5

I:\MEPC\63\23.doc
.5 P is the power of the main and auxiliary engines, measured in kW. The subscripts ME and AE refer to the main and auxiliary engine(s), respectively. The summation on i is for all engines with the number of engines (nME). (See diagram in appendix 1.)

.1 PME(i) is 75 per cent of the rated installed power (MCR *) for each main engine (i).

The influence of additional shaft power take off or shaft power take in is defined in the following paragraphs.

.2 Shaft generator

In case where shaft generator(s) are installed, PPTO(i) is 75 per cent of the rated electrical output power of each shaft generator.

For calculation of the effect of shaft generators two options are available:

Option 1:

.1 The maximum allowable deduction for the calculation of PME(i) is to be no more than PAE as defined in paragraph 2.5.6. For this case, PME(i) is calculated as:

PME(i) =) () (75 . 0 i PTOi ME PMCR
or

Option 2:

---CHUNK---

SOURCE: MEPC.212(63).pdf
Option 1:

.1 The maximum allowable deduction for the calculation of PME(i) is to be no more than PAE as defined in paragraph 2.5.6. For this case, PME(i) is calculated as:

PME(i) =) () (75 . 0 i PTOi ME PMCR
or

Option 2:

.2 Where an engine is installed with a higher rated power output than that which the propulsion system is limited to by verified technical means, then the value of PME(i) is 75 per cent of that limited power for determining the reference speed, Vref and for EEDI calculation.

* The value of MCR specified on the EI APP certificate should be used for calculation. If the main engines are not required to have an EIAPP certificate, the MCR on the nameplate should be used.

I:\MEPC\63\23.doc
The following figure gives guidance for determination of PME(i) :

.3 Shaft motor

In case where shaft motor(s) are installed, PPTI(i) is 75 per cent of

---CHUNK---

SOURCE: MEPC.212(63).pdf
Annex 8, page 6

I:\MEPC\63\23.doc
The following figure gives guidance for determination of PME(i) :

.3 Shaft motor

In case where shaft motor(s) are installed, PPTI(i) is 75 per cent of the rated power consumption of each shaft motor divided by the weighted average efficiency of the generator(s).

The propulsion power at which Vref is measured, is:

$$P_{Shaft\ i} = (P_{PTI\ i} + P_{ME}) / (\eta_{PTI\ i})$$

Where:

$$P_{PTI\ i} = P_{Shaft\ i} \cdot (\eta_{PTI\ i}) \cdot (\eta_{Gen})$$

$\eta_{PTI\ i}$ is the efficiency of each shaft motor installed
η_{Gen} is the weighted average efficiency of the generator(s)

Where the total propulsion power as defined above is higher than 75 per cent of the power the propulsion system is limited to by verified technical means, then 75 per cent of the limited power is to be used as the total propulsion power for determining the reference speed, Vref and for EEDI calculation.

I:\MEPC\63\23.doc
In case of combined PTI/PTO, the normal operational mode at sea

---CHUNK---

SOURCE: MEPC.212(63).pdf
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23
Annex 8, page 7

I:\MEPC\63\23.doc
In case of combined PTI/PTO, the normal operational mode at sea
will determine which of these to be used in the calculation.

Note: The shaft motor's chain efficiency may be taken into consideration to account for the energy losses in the equipment from the switchboard to the shaft motor, if the chain efficiency of the shaft motor is given in a verified document.

.4 Peff(i) is the output of the innovative mechanical energy efficient